

Automata Theory

Assignment #3

Due: May 9, 2008 (before Class)

1. (10 pts) Find a regular expression for the set $\{a^n b^m : n \geq 3, m \text{ is even}\}$.

Answer:

$$r = aaaa^*(bb)^*.$$

2. (10 pts) Give a simple verbal description of the language $L((aa)^*b(aa)^* + a(aa)^*ba(aa)^*)$

Answer: All strings are of the form w_1bw_2 , where w_1 and w_2 are composed of an even number of a 's, or w_1 and w_2 consists of an odd number of a 's.

3. (20 pts) Give regular expressions for the following languages.

(a) $L = \{w : n_a(w) \bmod 5 > 0\}$ on $\Sigma = \{a, b\}$.

(b) $L = \{w : n_a(w) \bmod 5 > 0\}$ on $\Sigma = \{a, b, c\}$.

Answer:

(a)

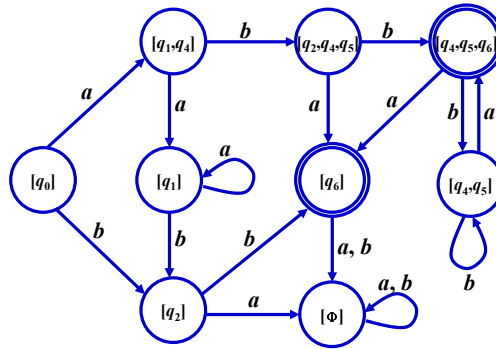
$$r = (b^*ab^* + b^*ab^*ab^* + b^*ab^*ab^*ab^* + b^*ab^*ab^*ab^*ab^*)(b^*ab^*ab^*ab^*ab^*)^*.$$

(b)

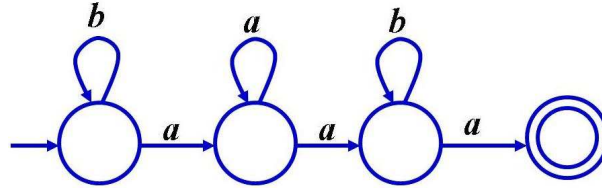
$$r = ((b+c)^*a(b+c)^* + (b+c)^*a(b+c)^*a(b+c)^* + (b+c)^*a(b+c)^*a(b+c)^*a(b+c)^* + (b+c)^*a(b+c)^*a(b+c)^*a(b+c)^*a(b+c)^*)((b+c)^*a(b+c)^*a(b+c)^*a(b+c)^*a(b+c)^*a(b+c)^*)^*.$$

4. (10 pts) Find a dfa that accepts $L(a^*bb) \cup L(ab^*ba)$.

Answer:



5. (10 pts) Find the regular expression for the language accepted by the following automata.



Answer:

$$r = b^*aa^*ab^*a.$$

6. (10 pts) Construct a right-linear grammar for the language $L((aab^*ab)^*)$.

Answer:

$$G = (V, T, S, P)$$

$$S \rightarrow aaA | \lambda$$

$$A \rightarrow bA | abS$$

7. (10 pts) Show that any regular grammar G for which $L(G) \neq \emptyset$ must have at least one production of the form

$$A \rightarrow x$$

where $A \in V$ and $x \in T^*$.

Answer: If there is no such rule, a sentence can never be derived.

8. (20 pts) Find regular grammars for the following languages on $\{a, b\}$.

(a) $L = \{w : (n_a(w) - n_b(w)) \bmod 3 \neq 1\}.$

(b) $L = \{w : |n_a(w) - n_b(w)| \text{ is odd } \}.$

Answer:

(a)

$$S \rightarrow A | AaAaA | AbAbA.$$

$$A \rightarrow aAb | bAa | \lambda.$$

(b)

$$S \rightarrow AaA | AbA.$$

$$A \rightarrow aAb | bAa | \lambda.$$

Note: Please use A4 papers to write up your homework solutions and do not forget to leave your name and student ID on it. If you fail to do so, you might get 0 on your homework grade. If you use more than one sheet for your homework solutions, please staple them before you hand in your homework.